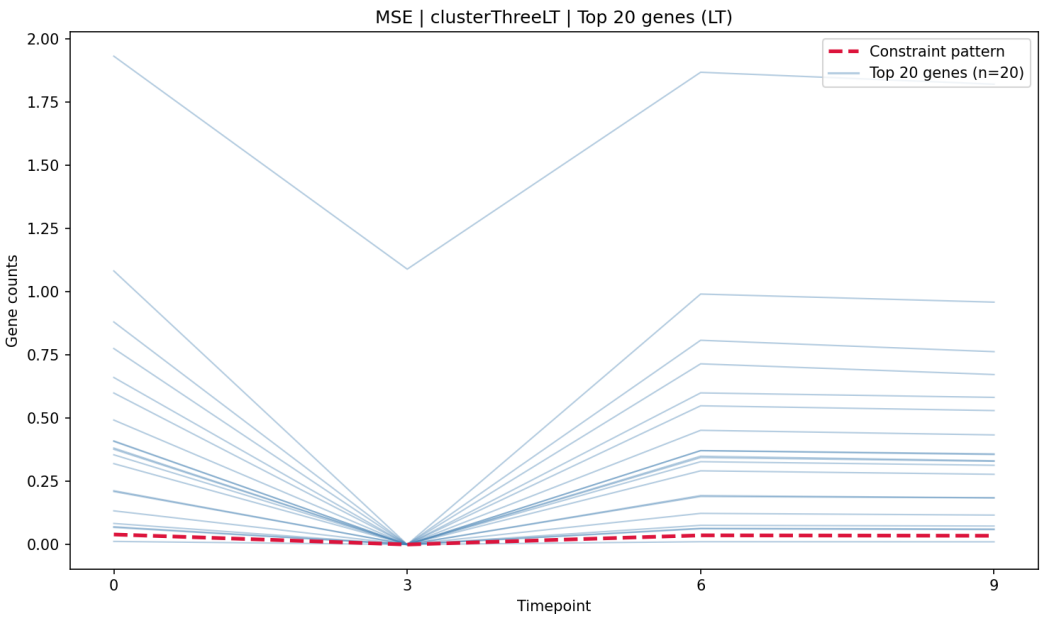
Method Gating & Visual Similarity

Week of April 16

Binghamton University

April 2026

Problem: Some Methods Pick the Wrong Genes

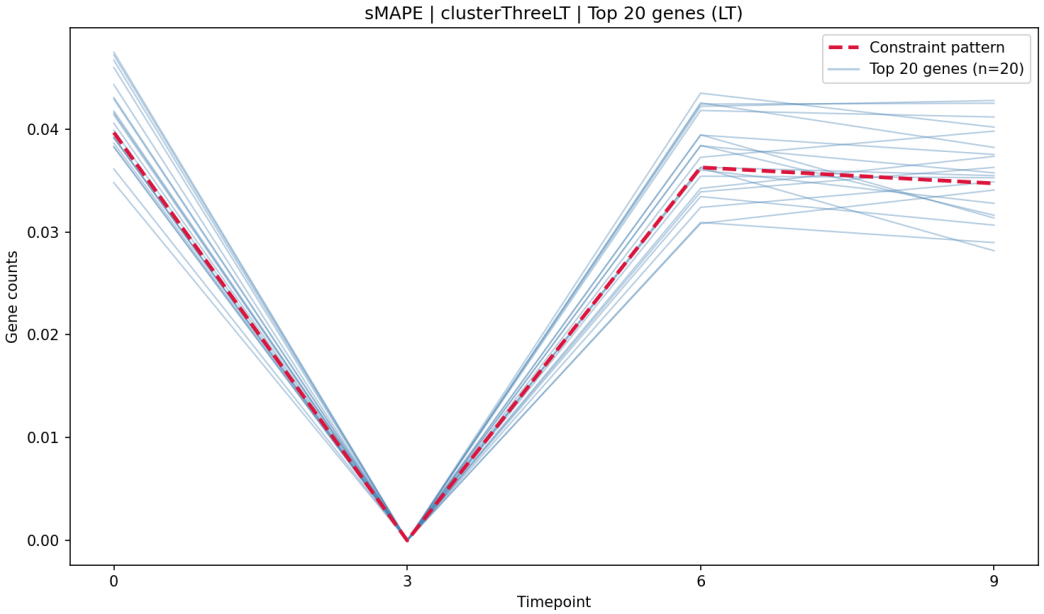


MSE on clusterThree LT

Top-20 genes are scattered far above the constraint pattern (red dashed).

MSE finds genes that are close on *normalized* data, but at completely different expression levels in reality.

Problem: Some Methods Pick the Wrong Genes



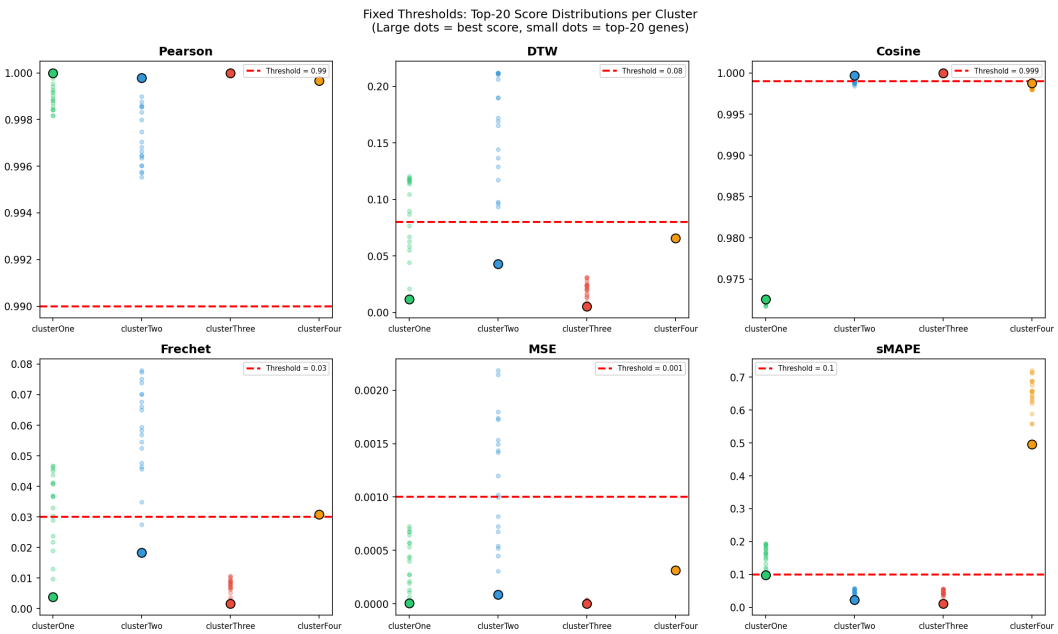
sMAPE on clusterThree LT

Top-20 genes tightly follow the constraint pattern.

Same cluster, same pattern — but only sMAPE finds the right genes.

Can we automatically detect when a method isn't working?

Attempt 1: Fixed Score Thresholds

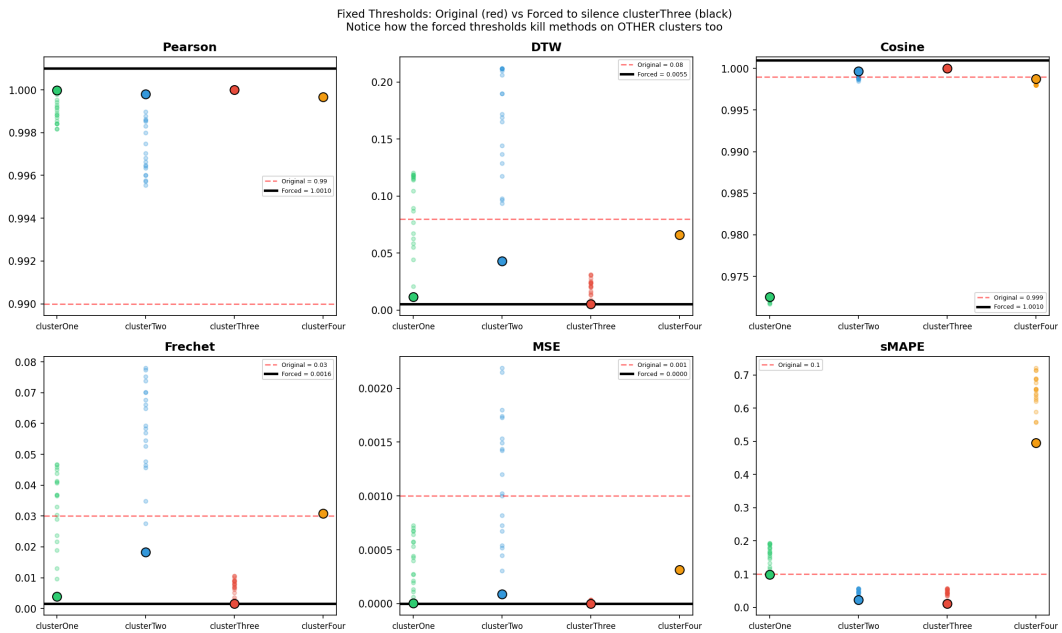


Idea: Set a quality floor on each metric's score. If a method's best genes don't meet it, the method can't vote.

Each subplot shows one method's top-20 scores across 4 clusters. The red line is the threshold.

Problem: The same method has very different score ranges across clusters — no single line works.

What If We Force the "Right" Answer?



Since sMAPE is the only method that visually works on clusterThree, what thresholds would we need to silence everything else there?

Black lines = forced thresholds.
Red lines = original thresholds.

The forced thresholds are so extreme they cut through *every* cluster's scores.

Forced Thresholds: Collateral Damage

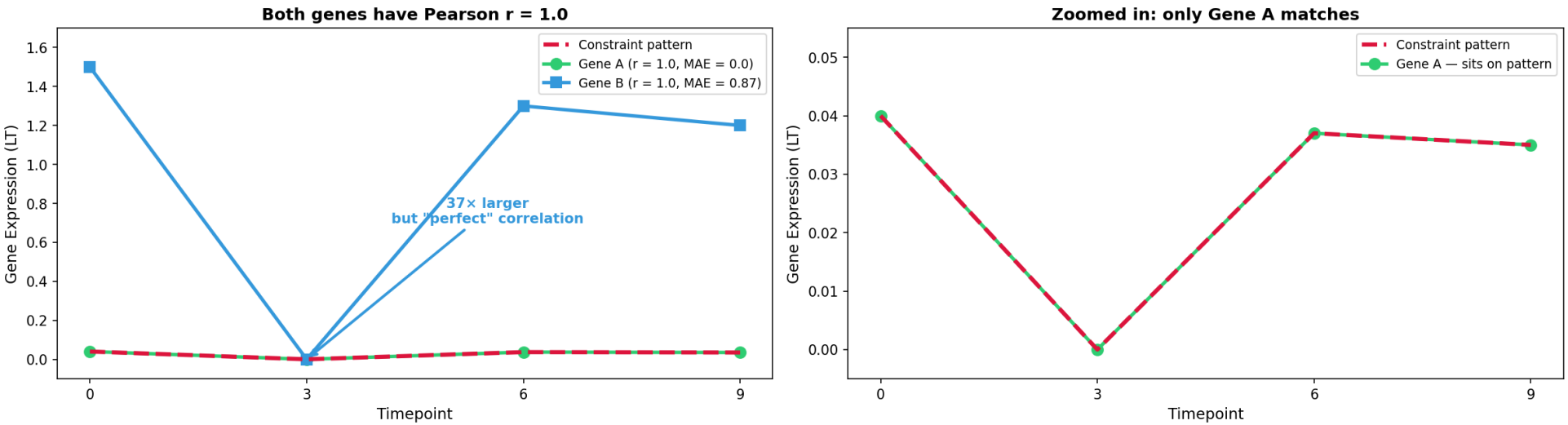
To silence non-sMAPE methods on clusterThree:

Method	**Original**	**Required**
Pearson	> 0.99	> 1.001 (impossible)
DTW	< 0.08	< 0.006
MSE	< 0.001	< 0.000001

Result: Every method silenced on every cluster except sMAPE — which only survives on 3 of 4 clusters.

Fixed thresholds can't work because the methods are actually *confident* on clusterThree (Pearson $= 1.0$) — just about the wrong genes.

Why Are the Methods So Confident on the Wrong Genes?



Gene B (blue) is 37× larger than the pattern, but both have Pearson $r = 1.0$.
On the plot Gene A (green) sits on the pattern; Gene B is nowhere near it.

Why Pearson = 1.0: Step-by-Step

Step 1: Pearson subtracts each series' mean before comparing. Gene B mean = 1.0, Pattern mean = 0.028

Step 2: After subtracting means, the residuals are:

	$t = 0$	$t = 3$	$t = 6$	$t = 9$
Gene B residual	$+0.50$	-1.00	$+0.30$	$+0.20$
Pattern residual	$+0.012$	-0.028	$+0.009$	$+0.007$
Same sign?	✓ $+/+$	✓ $-/-$	✓ $+/+$	✓ $+/+$

Step 3: Pearson multiplies the residuals at each timepoint:

	$t = 0$	$t = 3$	$t = 6$	$t = 9$	
product	$(+)(+)$ $= +0.006$	$(-)(-)$ $= +0.028$	$(+)(+)$ $= +0.003$	$(+)(+)$ $= +0.001$	sum $= +0.038$

Every product is positive because the signs always match.
The denominator just rescales to $[-1, 1]$: $r = 0.038 \,/\, 0.038 = \mathbf{1.0}$

Gene B is 37× larger, but the ups and downs match — so r is perfect.

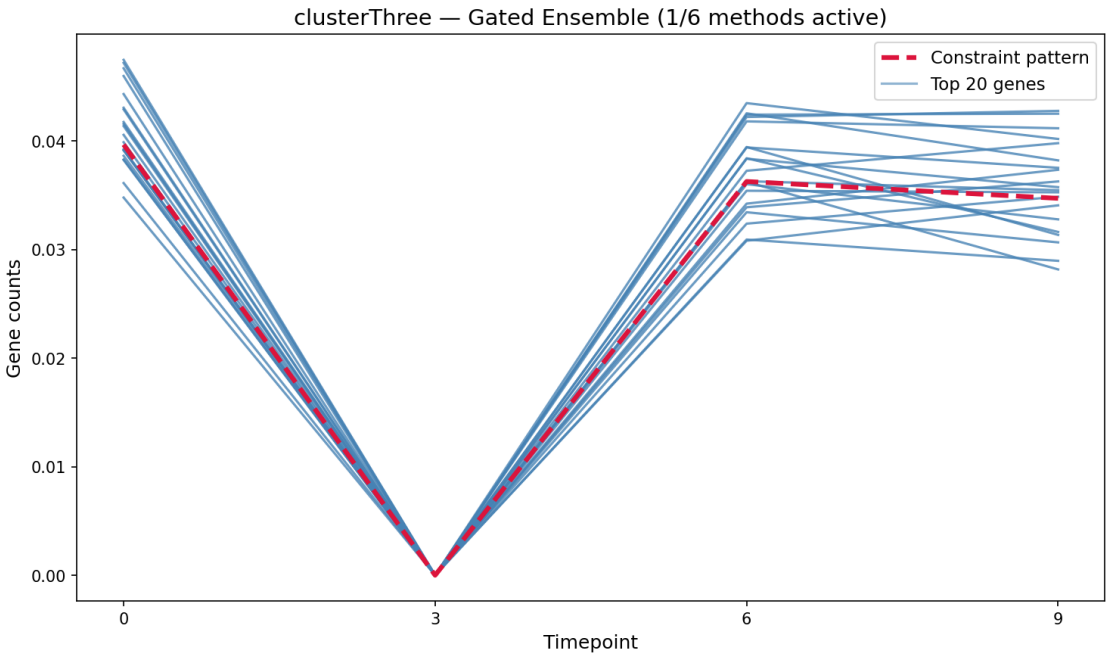
Attempt 2: Data-Driven Gating

- Fixed thresholds fail because the methods' own scores can't distinguish "confident and right" from "confident and wrong"
- **New idea:** check whether a method's top genes actually sit near the pattern in expression space
- For each method $\times$ cluster: average the top-20 genes into a centroid, then measure how far that centroid is from the pattern
- Normalize by pattern range so scores are comparable across clusters:

$$\text{Relative Centroid Error} = \frac{\text{MSE}(\bar{x}_{\text{top-20}},\ p)}{(\max p - \min p)^2}$$

- One universal cutoff applied everywhere $\rightarrow$ methods with high error are silenced

Gated Results: clusterThree (1/6 methods active — sMAPE only)

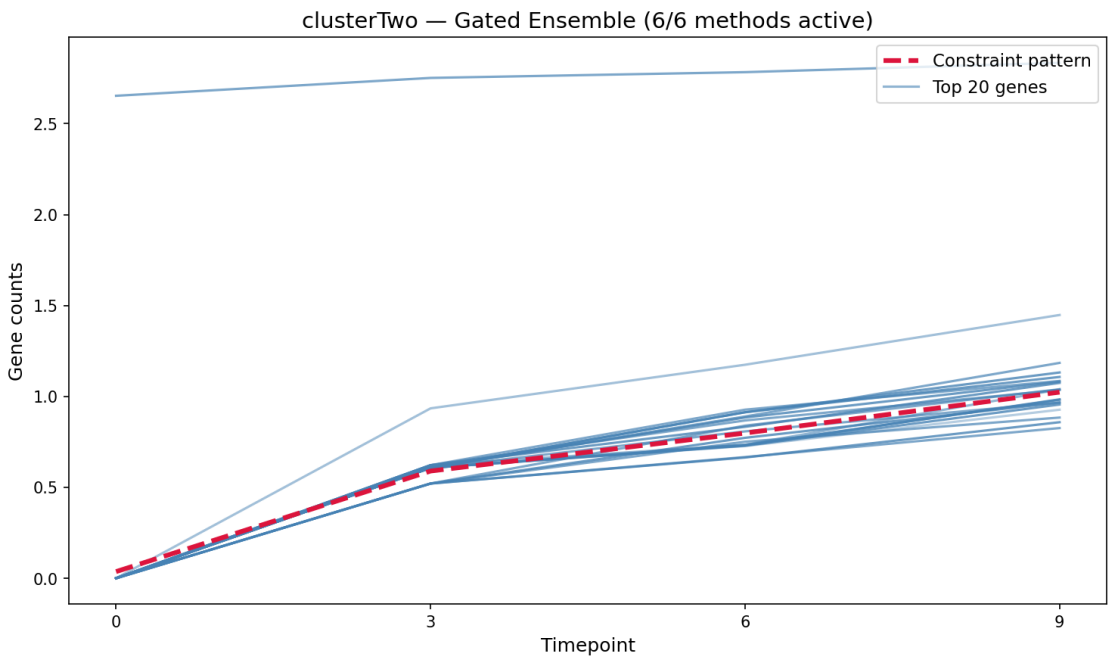


The gate correctly silences all methods except sMAPE on clusterThree.

sMAPE's centroid error: **0.001**
Other methods: 41–80

This matches exactly what we'd pick visually.

Gated Results: clusterTwo (6/6 methods active)

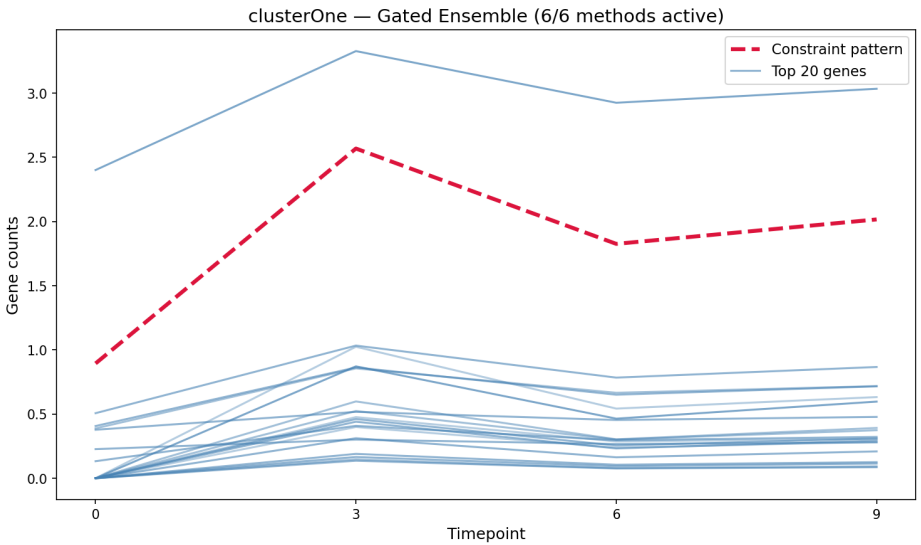


All 6 methods pass the gate — their top genes genuinely sit near the pattern.

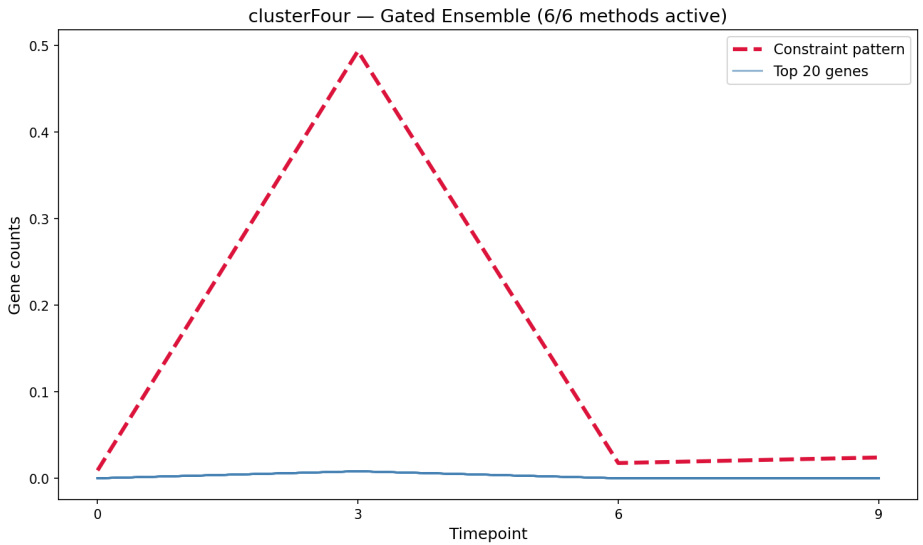
The gate correctly does *not* silence methods where they all work.

Same consensus genes as before (RAB14, SYNGR1, etc. at 6/6 unanimous).

But: Gated clusterOne and clusterFour Still Don't Fit



Genes follow the shape but sit below the pattern



6/6 active but genes are flat — nothing matches the spike

The gate checks if top genes are near the pattern *relative to its range* — so these pass because the centroid error is small *as a proportion*. But the genes are still far from the pattern in **absolute** terms. The root cause: all 6 methods operate on **normalized** data, which erases magnitude before scoring. Can we do better?

A Simpler Idea: Raw-Space MAE

All 6 methods work on **normalized** data —
so they find genes with the right *shape* but at any magnitude.

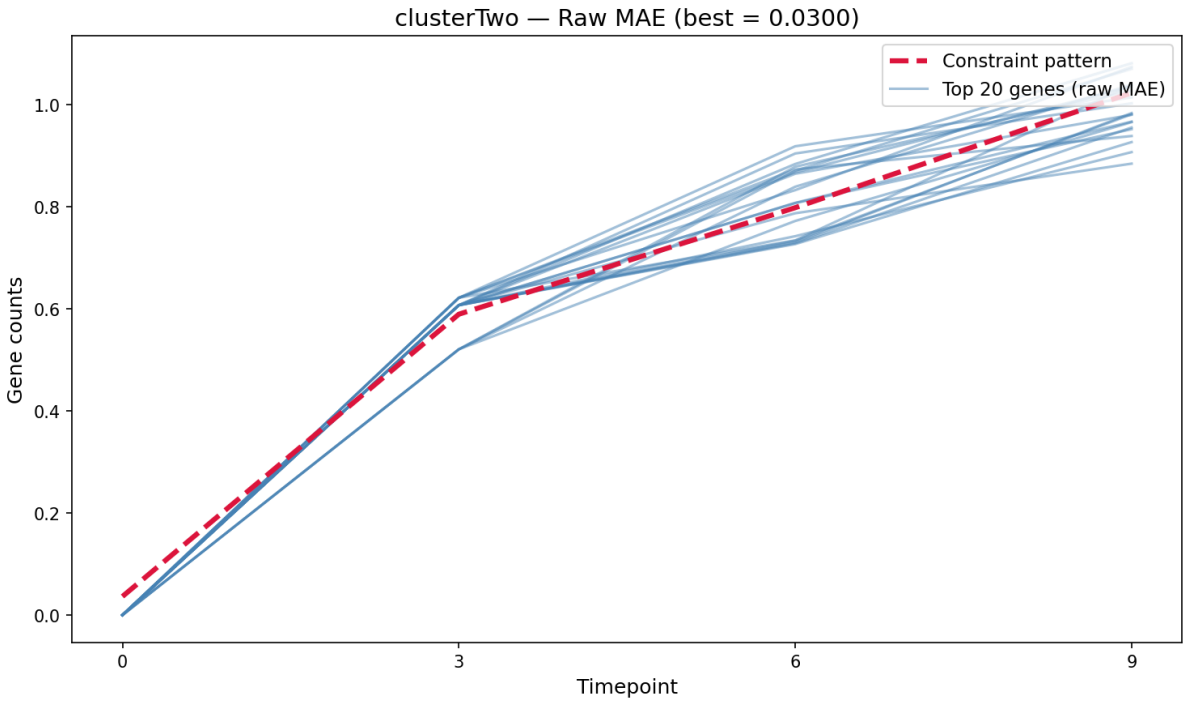
What if we skip normalization entirely?

$$\text{MAE} = \frac{1}{4} \sum_t |g_t - p_t|$$

No normalization. No correlation. No gating needed.

Just rank all 17,856 genes by how close they are
to the pattern **in the actual expression space**.

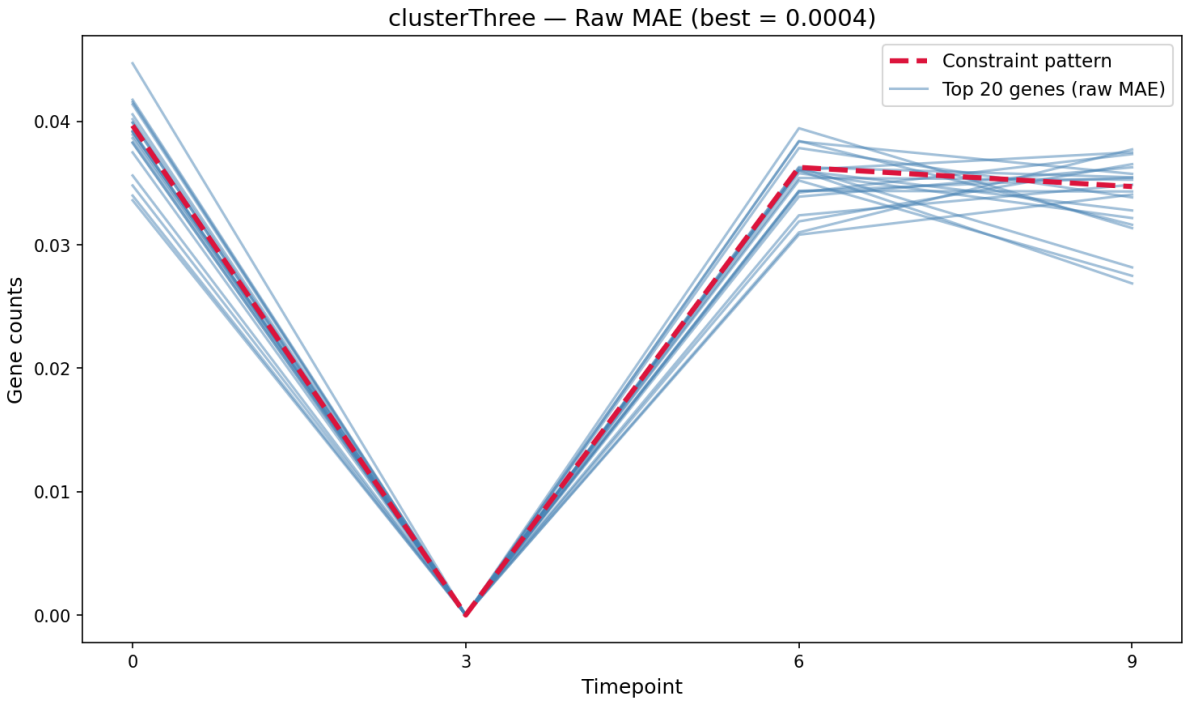
Raw MAE: clusterTwo



Genes tightly follow the steady-increase pattern.

8–12/20 overlap with all 6 original methods — confirming the existing consensus.

Raw MAE: clusterThree

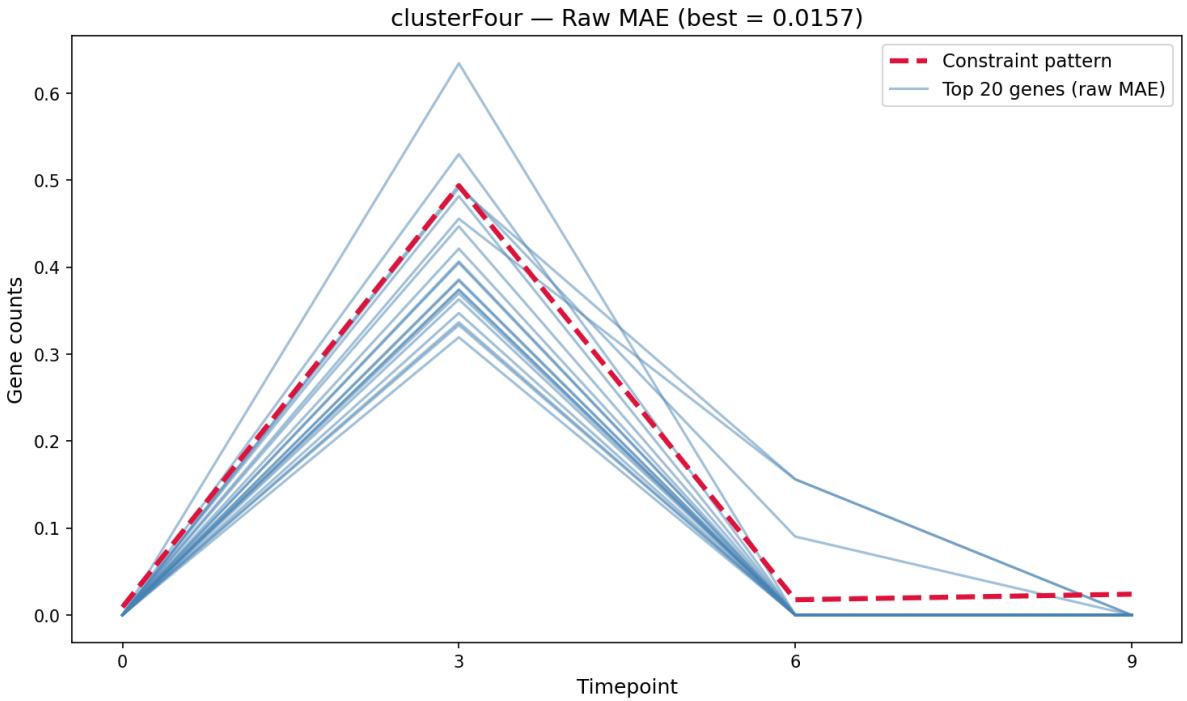


Genes sit right on the pattern.

11/20 overlap with sMAPE, 0/20 with all other methods.

Raw MAE independently confirms sMAPE was the only method finding the right genes.

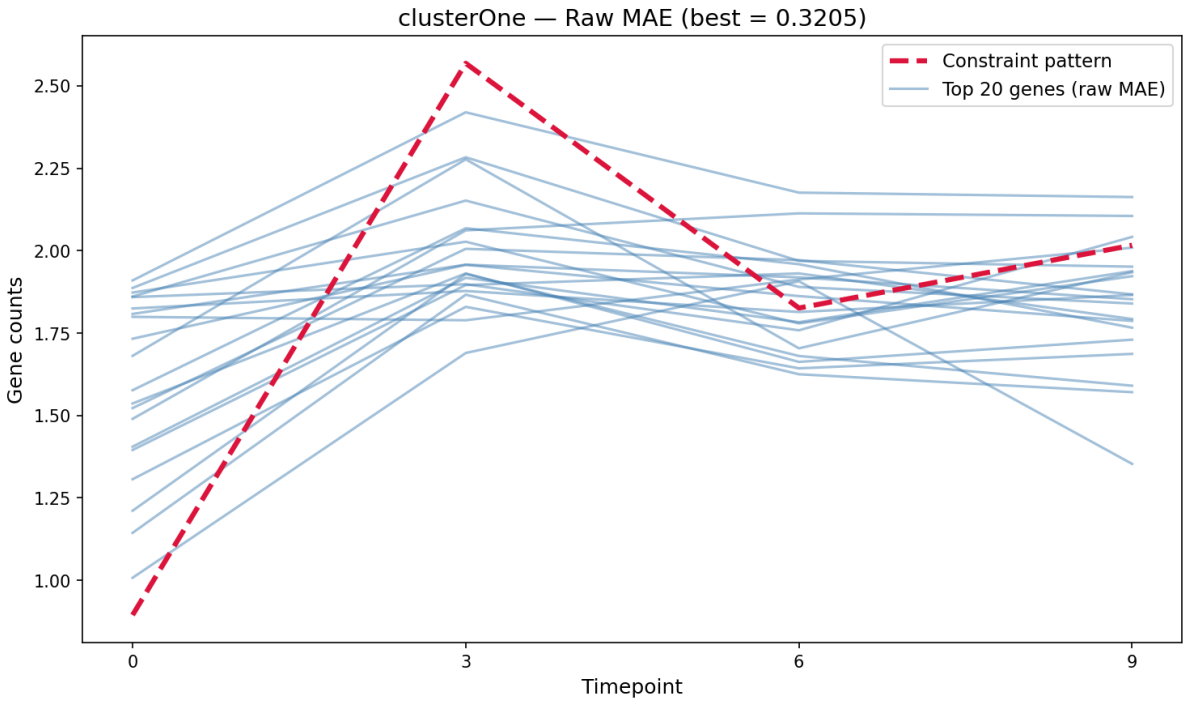
Raw MAE: clusterFour



For the first time, genes actually follow the spike pattern.

0/20 overlap with any original method — none of them found these genes because normalization erased the magnitude.

Raw MAE: clusterOne



The exception — genes at the pattern's magnitude don't follow the spike shape.

For this cluster, the shape-based methods may be more appropriate.

Raw MAE isn't always better — it depends on what the pattern looks like.

Discussion

- **Raw MAE** gives the best visual fit for clusters Two, Three, and Four
- **clusterOne** is the exception — genes at the pattern's magnitude don't follow its shape. Shape-based methods do better here.
- **Key insight:** all 6 original methods operate on normalized data, which erases magnitude. This is why they find genes with the right shape but at the wrong expression level.
- **Open question:** is matching the exact expression level biologically meaningful, or is shape similarity sufficient?
  If expression level matters $\rightarrow$ raw MAE on unnormalized data.
  If only the temporal shape matters $\rightarrow$ the normalized methods (Pearson, DTW, etc.) with gating may be more appropriate.